Sahrdaya College of Engineering and Technology, Kodakara

Answer Key

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
B.TECH DEGREE S4 (R,S) EXAMINATION JUNE 2023 (2019 SCHEME)

Course Code: EET202

Course Name: DC MACHINES & TRANSFORMERS

PART A

Answer all questions. Each question carries 3 marks

Q1 Explain why dummy coils are used in DC machines?

Ans

- The wave winding is possible only with particular number of conductor or coils.
- But sometimes the armature slots available in the armature winding do not meet the requirement of the winding. (It means that the available number of slots is more than the required number of conductors).
- Some slots are kept without armature winding in that case and dummy coil or coils are employed in that slots.
- The dummy coils are similar to other coils except their ends are cut, short and taped. They do not connect with the commutator bars.
- The dummy coils are simply **to provide mechanical balance for the armature**.
- As they do not connect with commutator bars they do not affect the electrical characteristics of the winding

Q2 Which are the different types of armature windings used in DC machines? Explain their differences and applications.

Ans Lap Windings and Wave Windings are the two different types of windings used in DC machines.

LAP WINDING	WAVE WINDING
• Finishing end of the first coil can be connected to the starting end of the next coil which starts from the same pole where the first coil started. • Armature windings are divided into several parallel paths • No. of parallel paths = no. of poles • Used for machines having large current & low voltage • Equalizer rings may be required	• Finishing end of the first coil can be connected to the starting end of the next coil which starts from the next similar pole where the first coil started • Only two parallel paths independent of number of poles • Used for machines with low current and high voltage • No need of equalizer rings

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- No. of brushes required is equal to no. of poles
- No need of dummy coils
- Used in Medium(50-500kW) and high power(>500kW) applications
- Commonly used for Current > 400A

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- Number of brushes required is always equal to two
- Dummy coils may be required
- Used for low power applications
- Commonly used for currents < 400A

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	Q3	Derive the emf equation of a DC generator.

Ans

Let,

φ : Flux per pole in Weber (Wb)
Z : Total number of armature conductors
P : Number of poles
A : Number of parallel paths
N : Speed of armature in rpm
E_g : EMF generated in volts

Flux cut by one conductor in one revolution of armature, dφ = Pφ
Time taken to complete one revolution, dt = 60/N sec

EMF generated in one conductor, $\frac{d\phi}{dt} = \frac{P\phi}{60/N} = \frac{PN\phi}{60}$

So, total EMF generated, E_g = (EMF per conductor) * number of conductors in one parallel path

$$E_g = \frac{PN\phi}{60} \times \frac{Z}{A} = \frac{Z\phi N}{60} \times \frac{P}{A}$$

	Q4	What are the conditions for voltage build up in self-excited DC generators?

Ans

Conditions to be satisfied for the voltage build up in a self excited generator are:

- Residual flux should be present
- Field connections should be proper
- Field circuit resistance should be less than the critical field resistance
- Speed of armature should be greater than critical speed.

	Q5	Why starters are used in DC motors?

Ans

$$E_b = \frac{Z\phi N}{60}\frac{P}{A} = K\phi\omega$$

$$V = E_b + I_a R_a$$

During the starting of DC motor,

→ Speed is zero
→ Back emf will be zero
→ Thus

		$$I_a = \frac{V}{R_a}$$ → Since armature resistance is very low, armature current will be very large. → This may lead to damage of armature windings and dip in supply voltage. → This can be avoided by the use of starters. → Starters introduce extra resistance in the armature path, so that armature current can be limited at safe range during starting.
	Q6	Explain the significance of back emf in DC motors.
	Ans	• Counter-Electromotive Force: When a DC motor operates, it generates a voltage known as back EMF. This voltage is induced in the armature windings due to the motor's rotation. It opposes the applied voltage (the supply voltage) and is referred to as a counter-electromotive force because it counteracts the flow of current into the motor. • Armature Current Control: Back EMF affects the amount of current flowing through the armature (the rotating part of the motor). Ohm's law states that current (I) is directly proportional to voltage (V) and inversely proportional to resistance (R) in a circuit (I = V / R). The back EMF, which opposes the supply voltage, reduces the effective voltage across the armature. As a result, it limits the armature current and helps prevent excessive current flow, which could otherwise damage the motor. This self-regulating property is especially important during startup or when the motor is subjected to sudden load changes. • Speed Regulation: Back EMF is directly proportional to the motor's speed. As the motor spins faster, the back EMF increases. This relationship is described by the equation: E_b = kω, where E_b is the back EMF, k is a constant, and ω is the angular velocity (speed) of the motor. By measuring the back EMF, you can indirectly determine the motor's speed. This property is valuable for maintaining precise control over the motor's speed in various applications.
	Q7	Define voltage regulation. What is the reason for negative voltage regulation in transformers?
	Ans	• When a transformer is loaded with a constant primary voltage, the secondary voltage decreases with increase in load. • This is because of its resistance and leakage reactance. • **The representation of the change in secondary terminal voltage from no-load to full-load as a percentage of full-load secondary terminal voltage is called Voltage Regulation.** • Lesser this value, better the transformer, because a good transformer should keep its secondary voltage as constant as possible under all load conditions. $$\%regulation = \frac{V_{2\_NL} - V_2}{V_2}$$ $$\%regulation = \frac{I_1 r_{01} \cos\phi \pm I_1 x_{01} \sin\phi}{V_1} \times 100$$ + : Lagging pf load – : Leading pf load

		For leading power factor loads, voltage regulation can be a positive value, negative value or zero.
	Q8	Derive the condition for maximum efficiency of a transformer.
	Ans	$Cu\ losses = I_1^{\,2} r_{01} = I_2^{\,2} r_{02}$ $Iron\ losses = W_i$ $Efficiency,\ \eta = \dfrac{Output}{Input} = \dfrac{Input - Losses}{Input}$ $= \dfrac{V_1 I_1 \cos\phi_1 - I_1^{\,2} r_{01} - W_i}{V_1 I_1 \cos\phi_1}$ $= \dfrac{V_1 I_1 \cos\phi_1}{V_1 I_1 \cos\phi_1} - \dfrac{I_1^{\,2} r_{01}}{V_1 I_1 \cos\phi_1} - \dfrac{W_i}{V_1 I_1 \cos\phi_1}$ $= 1 - \dfrac{I_1 r_{01}}{V_1 \cos\phi_1} - \dfrac{W_i}{V_1 I_1 \cos\phi_1}$ Differentiating with respect to I_1 $\dfrac{d\eta}{dI_1} = 0 - \dfrac{r_{01}}{V_1 \cos\phi_1} + \dfrac{W_i}{V_1 I_1^{\,2} \cos\phi_1}$ For Maximum Efficiency, $\dfrac{d\eta}{dI_1} = 0$ $\dfrac{r_{01}}{V_1 \cos\phi_1} = \dfrac{W_i}{V_1 I_1^{\,2} \cos\phi_1}$ $\boxed{I_1^{\,2} r_{01} = W_i}$ $\boxed{Cu\ loss = Core\ loss}$
	Q9	Explain the principle of on-load tap changing.
	Ans	On-load tap changing (OLTC) is a mechanism used in transformers to adjust the turns ratio and thereby regulate the output voltage while the transformer is under electrical load. Its principle can be summarized in three key points: • Continuous Voltage Regulation: The main principle of OLTC is to allow

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<tr><td></td><td></td><td>adjustments to the transformer's tap settings while it is actively supplying power to a load. This enables fine-tuning of the output voltage to meet changing load requirements or to compensate for voltage drops in the distribution system.

- Taps and Tap Changer: Transformers have multiple taps on their windings, which are like different connection points along the coil. The tap changer is a mechanism that can physically move the connection point between the primary and secondary windings to select a different tap. By changing the tap position, the turns ratio is altered, affecting the output voltage.
- Minimized Disruption: OLTC is designed to operate without interrupting the supply, ensuring a continuous and stable flow of electricity to the load. This minimizes disruptions and voltage fluctuations, making it suitable for applications where uninterrupted power supply is critical, such as in industrial settings or power distribution networks.
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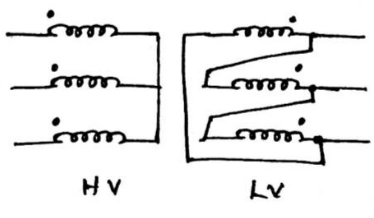
<tr><td></td><td>Q10</td><td>What is meant by vector groups? What does Yd11 mean?</td></tr>
<tr><td></td><td>Ans</td><td>

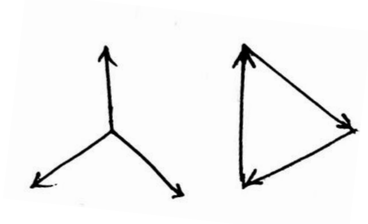
- In three phase transformers, there can be phase difference between primary and secondary voltages.
- This can be achieved by making slight changes in the winding configuration.
- Vector groups indicates the phase difference between primary and secondary voltages and thus the winding configuration.
- Various vector groups are Yy0, Yy6, Dd0, Dd6, Yd1, Yd11, Dyl and Dy11.
- In these vector groups, first letter indicates the type of connection of HV side, second letter indicates the type of connection of LV side and number indicates the phase shift.

Group 1: Zero phase displacement (Yy0, Dd0)
Group 2: 180 degree phase displacement (Yy6, Dd6)
Group 3: – 30 degree phase displacement (Yd1, Dy1)
Group 4: + 30 degree phase displacement (Yd11, Dy11)

<u>Yd11</u>
HV side is star connected whereas LV side is delta connected.
Voltage at LV side leads the voltage at HV side by 30°.



Connection Diagram


Phasor Diagram
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Answer any one full question from each module. Each question carries 14 marks

Module 1

Q11

a) Explain the terms (i) winding pitch (ii) front pitch and (iii) back pitch related to DC armature windings. Discuss how these values are selected in lap and wave windings. (6 Marks)

b) A 4 pole machine is wound with 564 conductors. The flux and speed are such that the average emf generated in each conductor is 2 V. The current in each conductor is 100A. Calculate the total current and the emf generated in the armature if the winding is (i) lap connected and (ii) wave connected. (8 Marks)

Ans

a)

- **Back pitch (Y_b):** Distance between two coil sides of the same coil.
- **Front pitch(Y_f):** Distance between 2 coil sides connected to the same commutator segments.
- **Winding pitch or Resultant pitch(Y_w):** Distance between similar coil sides of consecutive coils.

Design of Double Layer Simplex Lap Winding

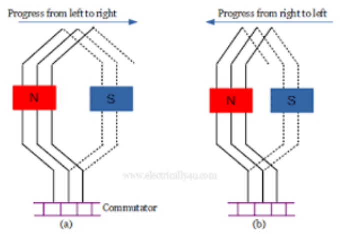
Back Pitch, $Y_b = \frac{Z}{P} \pm K$ (where K is an integer or fraction to make back pitch an odd integer)

Z : Number of Conductor
P : Number of Poles

Winding Pitch, $Y_w = \pm 2$

Front Pitch, $Y_f = Y_b - Y_w$

+ : for Progressive winding
– : for Retrogressive winding

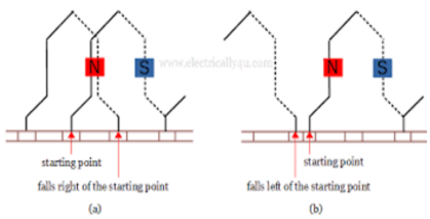


Design of Double Layer Wave Winding

Winding Pitch, $Y_w = \frac{Z \pm 2}{P/2}$

$Y_w = Y_b + Y_f$ (Y_b and Y_f should be odd)

Z : Number of Conductor
P : Number of Poles
+ : for Progressive winding
– : for Retrogressive winding



b) Given

		$P = 4$ $Z = 564$ $E_Z = 2$ V $I_Z = 100$ A (i) If lap connected A = p =4 Number of conductors in each parallele path = Z/4 = 564/4 = 141 Induced emf in each parallel path = 141 * 2 = 282 V Current in each parallel path = 100 A Therefore, Induced emf in armature = 282 V Armature current = 100 * 4 = 400 A (ii) If wave connected A = 2 Number of conductors in each parallele path = Z/2 = 564/2 = 282 Induced emf in each parallel path = 282 * 2 = 564 V Current in each parallel path = 100 A Therefore, Induced emf in armature = 564 V Armature current = 100 * 2 = 200 A
	Q12	Draw the developed winding diagram for a 12 slot, 4 pole simplex, progressive lap connected armature with 12 commutator segments. Prepare the winding table and mark the position of brushes. (14 Marks)
	Ans	Number of poles, P = 4 Number of armature conductors, Z = 2*number of commutator segments = 2*12 = 24 $Y_b = \frac{Z}{P} \pm K = \frac{24}{4} \pm K = 6 \pm K = 7$ $Y_w = 2$ $Y_f = Y_b - Y_w = 7 - 2 = 5$ *Winding Table:*

Back Connection	**Front Connection**
1 + 7 = 8	8 – 5 = 3
3 + 7 = 10	10 – 5 = 5
5 + 7 = 12	12 – 5 = 7
7 + 7 = 14	14 – 5 = 9
9 + 7 = 16	16 – 5 = 11
11 + 7 = 18	18 – 5 = 13
13 + 7 = 20	20 – 5 = 15
15 + 7 = 22	22 – 5 = 17
17 + 7 = 24	24 – 5 = 19
19 + 7 = 26 = 2	26 – 5 = 21
21 + 7 = 28 = 4	28 – 5 = 23
23 + 7 = 30 = 6	6 – 5 = 1

Module 2

Q13

a) What are the effects of armature reaction on the operation of dc machine? What are the remedial measures taken to counter the effects of armature reaction? (8 Marks)

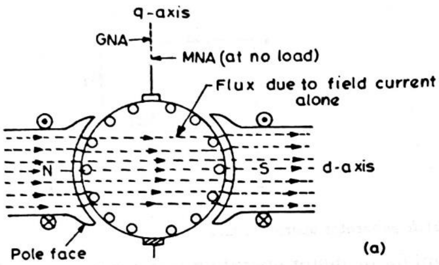
b) Explain with neat sketches, the process of commutation in DC machines. (6 Marks)

Ans

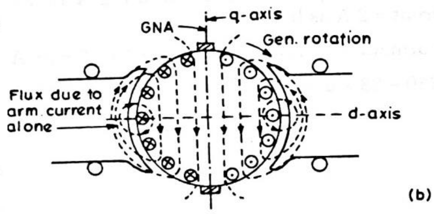
a) The action or effect of armature flux on main flux is known as armature reaction. There are mainly two effects:

Demagnetising effect: It is the effect of armature flux that weakens the main flux.

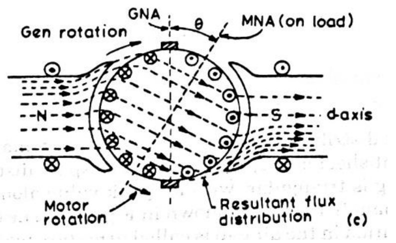
Cross magnetising effect: It is the effect of armature flux that distorts the main flux.



Main flux alone



Armature flux alone

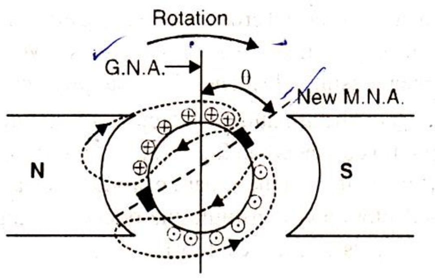


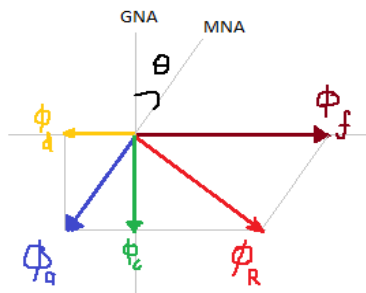
Combined effect of main flux and armature flux

Since the resultant direction of the flux in the armature is shifted, the magnetic neutral axis (MNA) get shifted from GNA.

For sparkles commutation process, brushes should be placed along MNA. Now the MNA is shifted, positions of brushes also should be changed.

After changing brush positions, armature flux distribution also will be changed.





- Now φ_a can be resolved in to 2 rectangular components φ_c and φ_d
- Component φ_d is in direct opposition to the main mmf. So it has a demagnetising effect on main flux and it is called **demagnetising (weakening) component of**

armature reaction.

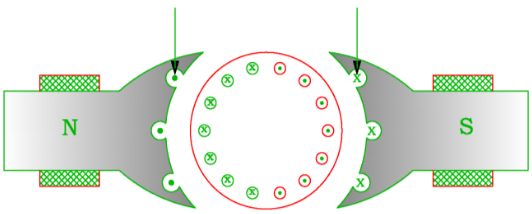
- The component φ_c is perpendicular to the main mmf. So it distort the main flux. Hence it is called **cross-magnetising (distorting) component of armature reaction.**
- The demagnetising effect leads to reduced generated voltage while cross-magnetising effect leads to sparking at the brushes.

<u>Methods to overcome armature reaction:</u>

❖ ***To overcome demagnetizing effect, number of field windings should be increased so that the strength of main flux can be increased.***

❖ ***To overcome cross magnetizing effect compensating windings need to be provided.***

- Compensating windings: Compensating windings are the auxiliary windings embedded in slots in the pole faces. They are connected in series with armature in a manner so that the direction of current through compensating conductors in any one pole face will be opposite to the current through the adjacent armature conductors.



b) Whenever armature conductors are alternately comes under different poles, direction of current changes. But in external circuit, we get unidirectional current. The process of converting bidirectional current in the armature conductors to unidirectional current in the external circuit by means of commutator segments and brushes is called **Commutation process**.

In other words, the reversal of current in the armature coil by means of commutator segments and brushes is known as **Commutation**.

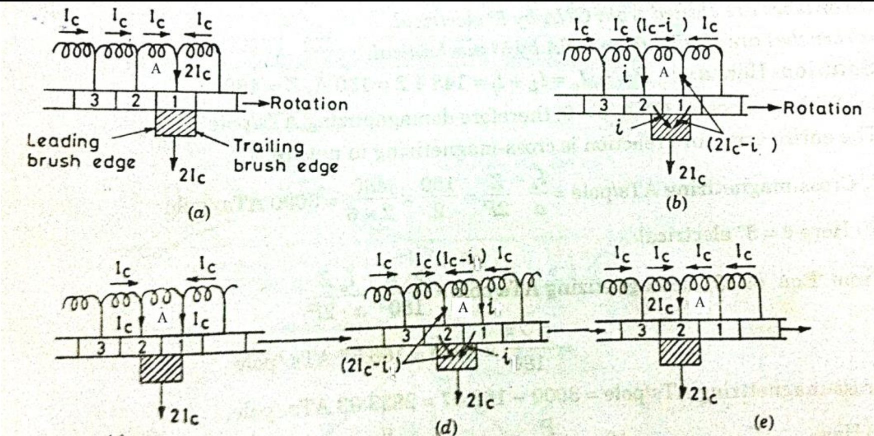


Fig. 4.24. Pertaining to the illustration of commutation process.

Figure (a)

- Brush is fully on commutator segment 1.
- Segment 1 conducts a current of $2I_c$ to the brush.
- I_c from coil A and I_c from the adjacent coil.

Figure (b)

- As armature rotates, brush makes contact with segment 2 and thus short circuits coil A.
- The brush is one-fourth on segment 2 and three-fourth on segment 1.
- Brush conducts a current of $2I_c$; i through segment 2 and $(2I_c - i)$ through segment 1.

Current in coil A is reduced to $(I_c - i)$ from I_c .

Figure (c)

- The brush is one-half on segment 1 and one-half on segment 2.
- Brush conducts a current of $2I_c$; I_c through segment 1 and I_c through segment 2.
- Current in coil A is reduced zero.

Figure (d)

- The brush is one-fourth on segment 1 and three-fourth on segment 2.
- Brush conducts a current of $2I_c$; i through segment 1 and $(2I_c - i)$ through segment 2.
- Coil A carries a current of $(I_c - i)$ in the opposite direction.

Figure (e)

- The brush is fully on segment 2.
- Brush conducts a current of $2I_c$; I_c through coil A and I_c from adjacent coil.
- Coil A carries a current of I_c but in the reverse direction.
- Thus coil A has undergone commutation.
- Each coil will undergo commutation in this way as it passes the brush axis.

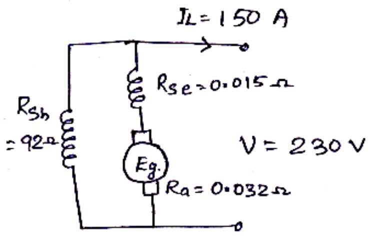
Q14 a) In a long-shunt compound generator, the terminal voltage is 230 V when the generator delivers 150 A. Determine (i) induced e.m.f. (ii) total power generated and (iii) power delivered to load. The shunt field, series field and armature resistance are

92 Ω , 0.015 Ω and 0.032 Ω respectively. (8 Marks)

b) What is meant by critical resistance of a DC shunt generator? How this is calculated from the OCC? (6 Marks)

Ans

a)



(i) Induced emf $E_g = V + I_a(R_a + R_{se})$

$$I_{sh} = \frac{V}{R_{sh}} = \frac{230}{92} = 2.5A$$

$$I_a = I_L + I_{sh} = 150 + 2.5 = 152.5\,A$$

$$E_g = 230 + 152.5\,(0.032 + 0.015)$$

$$= \underline{\underline{237.1675\ V}}$$

(ii) Total power generated, $P_g = E_g I_a$

$$= \underline{\underline{36.168\ KW}}$$

(iii) Power delivered to load, $P_o = V I_L$

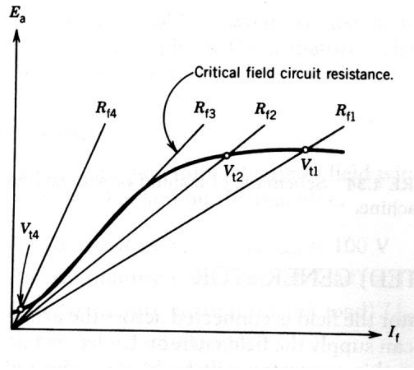
$$= 230 \times 150 = \underline{\underline{34.5\,KW}}$$

b)

Critical field resistance (R_c): It is the maximum field resistance above which there will be no voltage build up in the generator.

Critical resistance can be determined from OCC of DC machine as follows:

- Draw the OCC of DC machine at rated speed
- Draw a tangent to OCC in the linear region.
- Find the slope of the tangent
- Slope of tangent is nothing but the critical resistance of the DC machine at that particular speed.

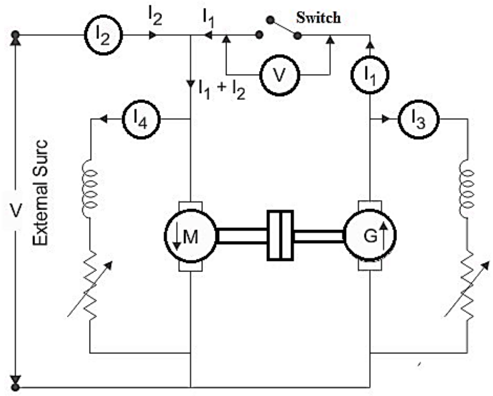


		Module 3
	Q15	a) Describe the principle of Hopkinson's test for testing of DC motor with the help of a neat circuit diagram. (8 Marks) b) Explain any two methods of speed control in a DC shunt motor. (6 Marks)

Ans

a) Hopkinson's Test

- Two identical dc machines are mechanically coupled.
- One of them runs as generator and other as a motor.
- Mechanical output of motor drives the generator and the electrical output of generator is supplying the electrical input for the motor.
- If there were no losses in the machines, they would have run without any external power supply.
- But, due to the losses associated with the machines, electrical output of generator will not be sufficient to run the motor and vice versa.
- These losses are supplied by the external DC source.



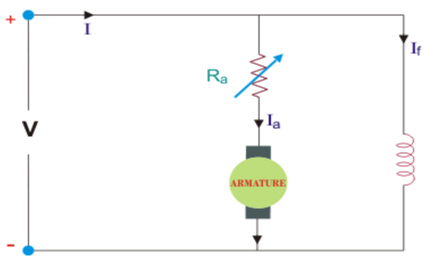
- Machine M is started as motor at rated voltage from the supply mains with switch is kept open.
- Speed is adjusted to rated speed.
- Machine M drives the Machine G as generator and voltage is read on voltmeter (V).
- Generator field resistance is adjusted to make the voltmeter reading as zero.

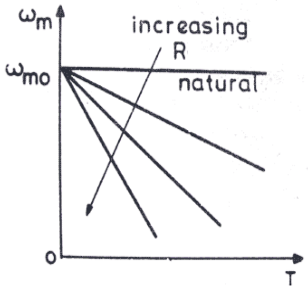
- When the voltmeter reads zero, switch can be closed.

b) Speed Control Techniques

Armature Control

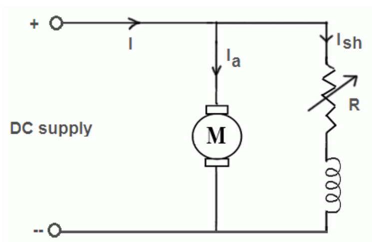
- Resistance of armature circuit is varied to control the speed
- As resistance increases, speed decreases and vice versa
- This method is used when speed control below base speed is required.
- A variable resistance is directly connected in series with the armature of dc motor.
- Speed is varied by wasting power in this external resistor.
- Since it is an inefficient method of speed control, this method is not commonly used.

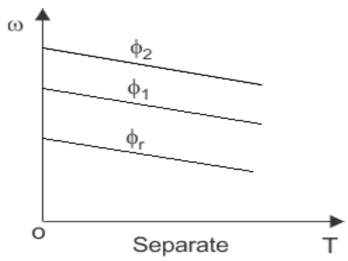




Field Control

- Flux is varied to control the speed
- Flux and speed are inversely proportional
- As flux increases, speed decreases and vice versa
- Since field flux is inversely proportional to the speed of motor, by varying field flux we can control the speed of motor.
- By decreasing the flux we can increase the motor speed and by increasing the flux we can decrease the speed.
- But, increasing of field flux is not preferred. It is because of the saturation of the magnetic core.
- So, by means of field flux control, speed of the motor is controlled only above base speed.



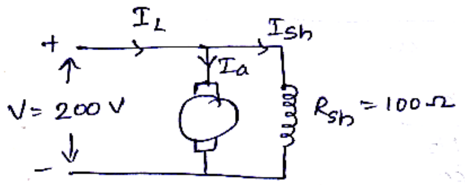


Q16 a) A 200-V DC shunt motor takes 4A at no-load when running at 700 rpm. The field resistance is 100 Ω. The resistance of armature at standstill gives a drop of 6 volts across armature terminals when 10 A were passed through it. Calculate (a) speed on rated load

(b) torque developed in N-m with rated load applied and (c) efficiency. The rated input to the motor is 8 kW. (10 Marks)

b) Derive the torque equation of a DC motor (4 Marks)

Ans a)



$$I_{sh} = \frac{V}{R_{sh}} = \frac{200}{100} = 2A$$

$$R_a = \frac{6}{10} = \underline{\underline{0.6\,\Omega}}$$

(a). Case 1 − No load condition.

$\cancel{I_{a1}}$ $I_{L1} = 4A$

$N_1 = 700$ rpm

$I_{a1} = I_{L1} - I_{sh} = 4 - 2 = \underline{\underline{2A}}$

$$E_{b1} = V - I_{a1}R_a$$
$$= 200 - (2 \times 0.6)$$
$$= \underline{\underline{198.8\,V}}$$

case 2 − rated condition

$P_{in} = VI_L = 8\,KW = 8000\,W$

$$I_{L2} = \frac{8000}{200} = 40\,A$$

$I_{a2} = I_{L2} - I_{sh} = 40 - 2 = \underline{\underline{38\,A}}$

$$E_{b2} = V - I_{a2}R_a = 200 - (38 \times 0.6)$$
$$= \underline{\underline{177.2\,V}}$$

$$\frac{E_{b1}}{E_{b2}} = \frac{N_1}{N_2}$$

$$N_2 = N_1 \times \frac{E_{b2}}{E_{b1}} = \underline{\underline{623.944\ rpm}}$$

(b) Torque developed $= \frac{\text{Power developed}}{\omega}$

$$= \frac{E_{b2} I_{a2}}{\frac{2\pi N_2}{60}}$$

$$= \underline{\underline{103.06 \text{ Nm}}}$$

(c) Here iron losses and mechanical losses are not given.

So, only electrical efficiency can be calculated

$$\eta_{ele.} = \frac{E_{b2} I_{a2}}{V I_L} = \frac{177.2 \times 38}{200 \times 40}$$

$$= \underline{\underline{84.17\%}}$$

b)

Let,

Z	: Number of armature conductors
B	: Magnetic field strength in Wb/m^2
I	: Current in armature conductor (I_a/A)
Φ	: Flux per pole in Wb
P	: Number of poles
A	: Number of parallel paths
r_a	: Radius of armature core
l	: Length of armature conductor

Force experienced on a single armature conductor, $F = BIlsin\theta = BIl$

Torque on each conductor, $\tau = BIl \times r_a$

Torque experienced on armature, $T = BIlr_a \times Z = Z\frac{\emptyset}{a}\frac{I_a}{A}lr_a$

Where 'a' is the cross-sectional area of flux path per pole and $a = \frac{2\pi r_a l}{P}$

Therefore, Torque developed, $T_e = \frac{Z\emptyset I_a P}{2\pi A} = 0.159 Z\emptyset I_a \frac{P}{A}$

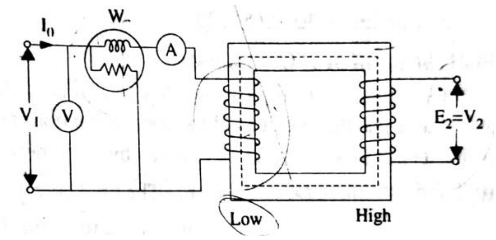
Module 4

Q17 a) With neat circuit diagram and relevant equations, explain how the efficiency of a transformer can be predetermined by conducting open circuit and short circuit tests. (10 Marks)

		b) Derive emf equation of a transformer. (4 Marks)
	Ans	a)

Open Circuit (OC) Test

- The purpose of this test is to determine the core losses and no-load current, which are helpful in finding X_m and R_c.
- It is performed on the low voltage (LV) side of the transformer.
- The other side (HV side) is left open.
- A voltmeter, wattmeter and an ammeter are connected in the LV side as shown in figure.
- Rated voltage is applied at the low voltage side.
- Then meter readings are noted.
- When rated voltage is applied at the LV side, rated flux is developed and thus core losses will occur.
- Since the secondary side is open circuited, transformer will draw only the no-load current, and it will be very small. (2% – 5% of rated current).
- Thus copper losses will be negligibly small at LV side and nil at HV side.
- Hence, wattmeter reading indicates the core losses of the transformer.
- Ammeter gives the no-load current.



Wattmeter reading, $W = V_1 I_e \cos\phi_o$

$$\cos\phi_o = \frac{W}{V_1 I_e} \quad \Rightarrow \quad \phi_o = \cos^{-1}\left(\frac{W}{V_1 I_e}\right)$$

$$I_m = I_e \sin\phi_o$$
$$I_c = I_e \cos\phi_o$$

$$X_m = \frac{V_1}{I_m}$$

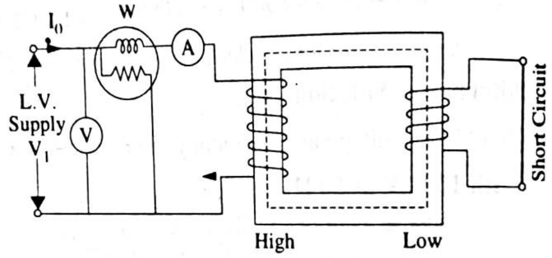
$$R_c = \frac{V_1}{I_c}$$

Short Circuit (SC) Test

- The purpose of this test is to determine the following:
  - → Cupper loss of the transformer at any load condition
  - → Voltage regulation
  - → Transformer impedance, leakage reactance and resistance.
- Low voltage side of the transformer is short circuited by a thick conductor.
- A low voltage (5% - 10% of the rated value) is applied at the HV side and is gradually increased till full load currents are flowing through the windings.
- Then meter readings are noted.
- Applied voltage is only a small percentage of the rated value, flux developed will be

small and hence core losses can be neglected.

- Thus, wattmeter gives the full-load copper losses of the transformer.



Wattmeter reading, $W = I_1^{\,2} r_{01}$

$$r_{01} = \frac{W}{I_1^{\,2}}$$

$$z_{01} = \frac{V_{sc}}{I_1}$$

$$x_{01} = \sqrt{z_{01}^{\;2} - r_{01}^{\;2}}$$

Efficiency = Output Power / Input Power = Output Power / (Output Power + Losses)

Total losses are calculated from SC and OC tests.

b)

When a sinusoidal voltage of frequency 'f' is applied across the primary windings, it produces a sinusoidal flux of same frequency.

Let, the flux at any instant can be given by $\phi = \phi_m \sin \omega t$

According to Faraday's Law,

Induced EMF, $e = -N \dfrac{d\phi}{dt}$

Thus, induced EMF in Primary winding

$$e_1 = -N_1 \frac{d\phi}{dt} = -N_1 \frac{d}{dt}(\phi_m \sin \omega t)$$
$$= -N_1 \phi_m \omega \cos \omega t$$
$$= -N_1 \phi_m \omega \sin(90 - \omega t)$$
$$= N_1 \phi_m 2\pi f \sin(\omega t - 90)$$

So,

RMS value of voltage induced in primary winding

$$E_1 = \frac{N_1 \phi_m 2\pi f}{\sqrt{2}}$$

$$\boxed{E_1 = 4.44 N_1 \phi_m f}$$

Similarly,

RMS value of voltage induced in secondary winding

$$\boxed{E_2 = 4.44 N_2 \phi_m f}$$

N_1 : Number of primary turns
N_2 : Number of secondary turns
f : Frequency of AC supply
ϕ_m: Maximum value of Flux in Wb

Q18 The low voltage winding of a 300-kVA, 11,000/2500-V, 50-Hz transformer has 190 turns and a resistance of 0.06 Ω. The high-voltage winding has 910 turns and a resistance of 1.6 Ω. When the LV winding is short-circuited, the full-load current is obtained with 540-V applied to the HV winding. Calculate (i) the equivalent resistance and leakage reactance as referred to HV side and (ii) Draw the equivalent circuit referred to primary. (14 Marks)

Ans

(i) $V_1 = 11000\ V$

$V_2 = 2500\ V$

$N_2 = 190 \qquad R_2 = 0.06\ \Omega$

$N_1 = 910 \qquad R_1 = 1.6\ \Omega$

During short circuit test, (HV)

$V_{sc} = 540$

$I_{sc} = \frac{300 \times 10^3}{11000} = 27.2727 A$

$Z_{01} = \frac{540}{27.27} = 19.8\ \Omega$

$K = \frac{V_2}{V_1} = \frac{2500}{11000} = 5/22$

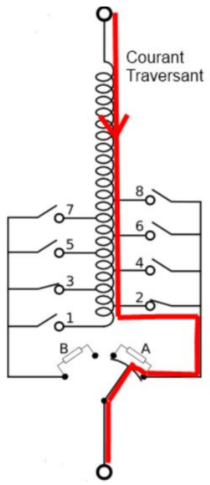
		$R_2' = \frac{R_2}{K^2} = \underline{1.1616\,\Omega}$ $R_{01} = R_1 + R_2' = \underline{\underline{2.7616\,\Omega}}$ $X_{01} = \sqrt{Z_{01}^2 - R_{01}^2} = \underline{\underline{19.61\,\Omega}}$
		Module 5
	Q19	a) Explain with neat sketches, the principle of online tap changing. (10 Marks) b) Explain how the power transfer from primary to secondary of an autotransformer takes place partly by conduction and partly by induction. (4 Marks)
	Ans	a) Online tap changing (OLTC) is a mechanism used in transformers to adjust the turns ratio and thereby regulate the output voltage while the transformer is actively supplying power to a load. It ensures that the voltage supplied to the load remains within acceptable limits, compensating for variations in load conditions and voltage drops in the distribution system. Here's an explanation of the principle of online tap changing with a simplified sketch: **Components and Operation:** Transformer Windings: A typical transformer has two windings, a primary winding and a secondary winding. The primary winding is connected to the high-voltage source (HV), while the secondary winding is connected to the load and delivers low-voltage power. Tap Changer: Inside the transformer tank, there is a tap changer mechanism that includes a set of tap leads connected to different points along the primary winding. These tap leads can be physically adjusted to change the point at which the primary winding is connected to the secondary winding. Motor Drive: The tap changer mechanism is controlled by a motor drive system, which can be operated manually or automatically based on a voltage control system. In an automatic system, sensors continuously monitor the output voltage, and when it deviates from the desired setpoint, the motor drive adjusts the tap position. **Operation Steps:** Initial Position: The tap changer is initially set to a particular tap position, which determines the turns ratio of the transformer. The turns ratio affects the output voltage. Load Variations: As the load on the transformer varies or when there are voltage fluctuations in the grid, the output voltage may deviate from the desired level. Voltage Monitoring: Sensors continuously monitor the output voltage. If the voltage

goes above or below the desired range, it triggers a tap-changing action.

Tap Adjustment: The motor drive system adjusts the position of the tap changer. By moving the tap leads to a different position along the primary winding, the turns ratio is altered, and the output voltage is adjusted accordingly.

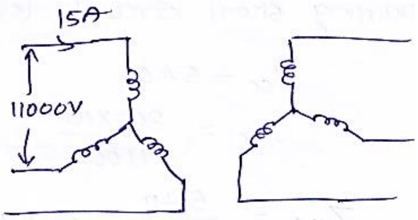
Steady-State Operation: Once the new tap position is set, the transformer continues to supply power to the load with the corrected output voltage. This process happens seamlessly and without interrupting the power supply, making it an online operation.

Online tap changing ensures that the output voltage remains stable and within acceptable limits, optimizing the performance of the transformer and ensuring a reliable power supply to the load even under varying conditions

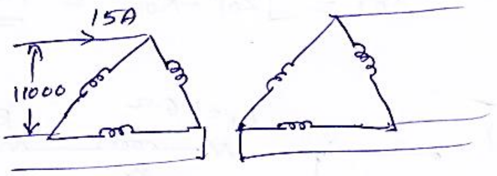


b) An autotransformer is a type of transformer that uses a single winding for both the primary and secondary coils, with a portion of the winding shared between them. Power transfer in an autotransformer occurs partly by conduction and partly by induction. Conduction involves the direct flow of electrical current through the common winding, while induction involves the electromagnetic coupling between the primary and secondary windings, which induces voltage and facilitates power transfer. This combination of conduction and induction allows for efficient voltage transformation in autotransformers while minimizing the size and weight compared to conventional transformers with separate primary and secondary windings.
Conduction (Direct Current Path):

In an autotransformer, there is a portion of the winding that is common to both the primary and secondary sides. This shared portion is called the common winding or the series winding.
When an AC voltage is applied to the primary side of the autotransformer, current flows through the common winding because it is in series with both the primary and secondary coils.
The current in the common winding creates a magnetic field around it due to Ampere's law, similar to how it works in a regular transformer.
This magnetic field couples with the secondary winding, inducing a voltage in the

		secondary winding by electromagnetic induction. The induced voltage in the secondary winding allows power transfer from the primary to the secondary through conduction, as the electrical energy flows directly through the common winding and is coupled to the secondary winding. Induction (Electromagnetic Coupling): In addition to conduction, autotransformers also utilize electromagnetic induction principles. When AC voltage is applied to the primary winding, it produces a varying magnetic field in the transformer core. This varying magnetic field cuts across the entire winding, including the common winding and the secondary winding. According to Faraday's law of electromagnetic induction, a changing magnetic field induces a voltage in any nearby conductor. As a result of the changing magnetic field produced by the primary winding, a voltage is induced not only in the common winding but also in the secondary winding, due to their electromagnetic coupling. This induced voltage in the secondary winding also allows power transfer from the primary to the secondary through electromagnetic induction.
	Q20	A bank of three single- phase transformers is connected to 11000V supply and takes 15A. If the ratio of turns per phase is 10, calculate the primary and secondary phase currents, voltages and the power output for the following connections (i) stat- star (ii) delta- delta (iii) star-delta (iv) delta- star.
	Ans	$V_{L1} = 11000\,V$ $I_{L1} = 15\,A$ $\frac{N_1}{N_2} = 10$ (i) Star – Star  $V_{Ph1} = \frac{11000}{\sqrt{3}} = \underline{6.350\,kV}$ $I_{Ph1} = \underline{15A}$ $V_{Ph2} = \frac{N_2}{N_1} \times V_{Ph1}$ $= \frac{1}{10} \times 6.350 = \underline{635\,V}$ $I_{Ph2} = \frac{N_1}{N_2} I_{Ph1} = 10 \times 15 = \underline{150A}$ $P_{out} = 3 V_{Ph2} I_{Ph2} = \underline{285.75\,KVA}$

(ii) delta – delta.



$I_{Ph_1} = \frac{15}{\sqrt{3}} = 8.66\,A$

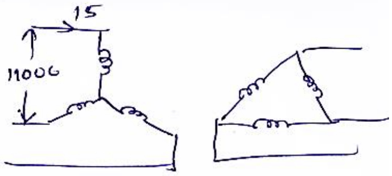
$V_{Ph_1} = 11000\,V$

$V_{Ph_2} = \frac{N_2}{N_1} V_{Ph_1} = \frac{1}{10} \times 11000 = 1100\,V$

$I_{Ph_2} = 10 \times I_{Ph_1} = 86.6\,A$

$P_{out} = 3 V_{Ph_2} I_{Ph_2} = 285.78\ KVA$

(iii) star – delta



$V_{Ph_1} = \frac{11000}{\sqrt{3}} = 6350\,V$

$I_{Ph_1} = 15$

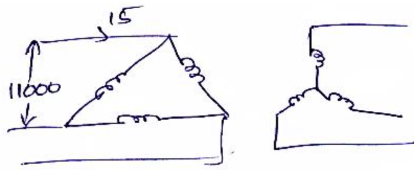
$V_{Ph_2} = \frac{N_2}{N_1} V_{Ph_1}$

$= \frac{1}{10} \times 6350 = 635\,V$

$I_{Ph_2} = \frac{N_1}{N_2} I_{Ph_1} = \frac{1}{10}\ 10 \times 15 = 150\,A$

$P_{out} = 285.75\ KVA$

(iv) delta – star



$V_{Ph_1} = 11000\,V$

$I_{Ph_1} = \frac{15}{\sqrt{3}} = 8.66\,A$

$V_{Ph_2} = \frac{N_2}{N_1} \times V_{Ph_1} = \frac{1}{10} \times 11000$

$= 1100\,V$

$I_{Ph_2} = \frac{N_1}{N_2} I_{Ph_1} = 10 \times 8.66 = 86.6\,A$

$P_{out} = 285.78\ KVA$